

Smart Polymer Composites with Multifunctional Capabilities Integrating Electroactive Polymers Conductive Nanofillers and Flexible Electronics for Advanced Sensing and Actuation Systems

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Abstract

Smart polymer composites have gained significant attention to their ability to integrate polymer matrices with conductive nanofillers, offering tunable electrical, mechanical, and electroactive properties. These composites are highly responsive to external stimuli such as electrical fields, mechanical stress, and temperature variations, making them ideal for applications in flexible electronics, soft robotics, and adaptive sensing systems. This research investigates the effect of nanofiller dispersion on the performance of polymer composites, optimizing nanofiller concentration to enhance electrical conductivity, mechanical strength, and flexibility. A one-way ANOVA was used to analyze the significance of material composition, particularly nanofiller concentration and polymer type, on composite properties. The long-term environmental stability of EAPs, such as polypyrrole (PPy) and polyvinylidene fluoride (PVDF), was examined to improve their reliability. The integration of flexible electronics through conductive inks and screen printing was explored to enhance the mechanical compliance and durability of these composites in wearable devices and robotics. ML models, including Random Forest, Support Vector Machines (SVM), and Neural Networks, are employed for multi-property optimization, predicting and improving composite performance. The study also addresses the scalability and cost-effectiveness of fabrication techniques to ensure the commercial viability of these advanced materials. The potential for incorporating biodegradable or environmentally friendly materials into smart polymer composites was explored to meet the growing demand for sustainable technologies. This research aims to advance the design, optimization, and application of smart polymer composites, paving the way for their widespread use in next-generation electronic and robotic systems.

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